Quantum Computing, Quantum Machine Learning and Quantum Information Theories

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Plans for the week of March 17-21, 2025

1. Lipkin model for VQE quantum computing
2. Discussion and work on project 1

Reading recommendation: Hundt's chapter 3 and 4 contain many useful hints for computing properties of Pauli matrices. Similarly, section 6.11 has many details of relevance for project 1.

More background material

For the Lipkin model, we recommend strongly the work of LaRose and collaborators, see https://journals.aps.org/prc/abstract/10.1103/PhysRevC.106.024319, the article is also available at https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/articles/PhysRevC.106.024319.pdf. See in particular section 3.

For codes, feel free to be inspired and/or reuse the codes at https://github.com/CompPhysics/QuantumComputingMachineLearning/tree/gh-pages/doc/Programs/LipkinModel.

Hamiltonians and project 1

As an initial test, we consider a simply 2×2 real Hamiltonian consistend of a diagonal part H_0 and off-diagonal part H_I, playing the roles of a non-interactive one-body and interactive two-body part respectively. Defined through their matrix elements, we express them in the Pauli basis $|0\rangle$ and $|1\rangle$

$$H = H_0 + H_I$$

$$H_0 = \begin{bmatrix} E_1 & 0 \\ 0 & E_2 \end{bmatrix}, \qquad H_I = \lambda \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}$$

Where $\lambda \in [0, 1]$ is a coupling constant parameterizing the strength of the interaction.

Rewriting in terms of Pauli matrices

Defining

$$E_+ = \frac{E_1 + E_2}{2}, \qquad E_- = \frac{E_1 - E_2}{2}$$

we see that by combining the identity and Z Pauli matrix, this can be expressed as

$$H_0 = E_+ I + E_- Z$$

For H_1 we use the same trick to fill the diagonal, defining $V_+ = (V_{11} + V_{22})/2, V_- = (V_{11} - V_{22})/2$. From the hermiticity requirements of H, we note that $V_{12} = V_{21} \equiv V_o$, which simplifies the problem to a simple X. This gives

$$H_I = V_+ I + V_- Z + V_o X$$

Measurement basis

For this system we note that the Pauli X matrix can be rewritten in terms of the Hadamard matrices and the Pauli Z matrix, that is

$$X = HZH.$$

Second Hamiltonian matrix problems

The second case is defined as a 4×4 real Hamiltonian. This can be viewed as two composite systems where each system is a two-level system. In the product basis $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$ we define the one-body part as

$$H_0|ij\rangle = \epsilon_{ij}|ij\rangle,$$

with a two-body interaction defined as

$$\begin{aligned} H_I &= H_x X \otimes X + H_z Z \otimes Z \\ &= \begin{bmatrix} H_z & 0 & 0 & H_z \\ 0 & -H_z & H_x & 0 \\ 0 & H_x & -H_z & 0 \\ H_x & 0 & 0 & H_z \end{bmatrix}. \end{aligned}$$

Here H_x and H_z are coupling constants.

Rewriting

For the 4×4 case, the interacting part of the Hamiltonian is already written in terms of Pauli matrices. On the other hand, we need to be rewrite the diagonal term. We define

$$\epsilon_{\pm 0} = \frac{\epsilon_{00} \pm \epsilon_{01}}{2}, \qquad \epsilon_{\pm 1} = \frac{\epsilon_{10} \pm \epsilon_{11}}{2}.$$

We note that the energies ϵ_{00} and ϵ_{01} can be repeated on the diagonal. This is also the case for ϵ_{10} and ϵ_{11}

$$D_0 = \epsilon_{+0} I \otimes I + \epsilon_{-0} I \otimes Z,$$
$$D_1 = \epsilon_{+0} I \otimes I + \epsilon_{-1} I \otimes Z.$$

Further manipulations

Using the Pauli Z matrix and the identity matrix I we define

$$P_{\pm} = \frac{1}{2}(I \pm Z),$$

which we use to project out the first and last to two elements the of 4×4 matrix

$$P_0 = P_+ \otimes I, \qquad P_0 = P_- \otimes I.$$

Adding D_0 and D_1, we get

$$H_0 = P_0 D_0 + P_1 D_1 = \alpha_+ I \otimes I + \alpha_- Z \otimes I + \beta_+ I \otimes Z + \beta_- Z \otimes Z,$$

where we have defined

$$\alpha_{\pm} = \frac{\epsilon_{+0} \pm \epsilon_{+1}}{2}, \quad \beta_{\pm} = \frac{\epsilon_{-0} \pm \epsilon_{+1}}{2}$$

The Lipkin model

Last week we introduced the Lipkin model which could be rewritten in terms of the quasipin operators as

$$
\begin{aligned}
H_0 &= \epsilon J_z, \\
H_1 &= \frac{1}{2} V (J_+^2 + J_-^2), \\
H_2 &= \frac{1}{2} W (J_+ J_- + J_- J_+ - N),
\end{aligned}
$$

Quasispin operators

These quasi spin operators obey the normal spin commutator relations

$$[J_z, J_\pm] = \pm J_\pm, \qquad [J_+, J_-] = 2J_z,$$
$$[J^2, J_\pm] = 0, \qquad [J^2, J_z] = 0,$$

in addition to commuting with the number operator

$$[N, J_z] = [N, J_\pm] = [N, J^2] = 0.$$

Using these relations we can show that the Hamiltonian (a product of quasi spin operators and the number operator) also commutes with J^2, that is $[H, J^2] = 0$. This means that H and J^2 have a shared eigenbasis, with J being a so-called **good** quantum number.

System to diagonalize by traditional methods

Using spin-eigenstates as the Hamiltonian basis, we define states through the normal approach $|J, J_z\rangle$ with J and J_z as spin and spin-projections, respectively. The states $J_z = \pm J$ are the easiest to construct, corresponding to a single level being completely filled. States in between can then be found using the quasi-spin ladder operators following

$$J_{\pm}|J, J_z\rangle = \sqrt{J(J+1) - J_z(J_z \pm 1)}|J, J_z \pm 1\rangle.$$

Using this basis for the quasi-spin Hamiltonian, we can rewrite the explicit matrix $H_{J_z, J_z'} = \langle J, J_z|H|J, J_z'\rangle$ can be constructed.

The $J = 1$ case

For $N = 2$ particles, we have the triplet $J = 1$, with three possible projection $J_z = 0, \pm 1$. This gives the following matrix

$$H = \begin{bmatrix} -\epsilon & 0 & V \\ 0 & W & 0 \\ V & 0 & \epsilon \end{bmatrix}.$$

By solving the eigenvalue problem, the ground state energy can be exactly found.

The $J = 2$ case

Similarly, for the $N = 4$ particles case we have $J = 2$ with five possible projections $J_z = 0, \pm 1, \pm 2$. This gives us the Hamiltonian matrix

$$H = \begin{bmatrix} -2\epsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\epsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \epsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\epsilon \end{bmatrix}.$$

More rewriting

Following the work of LaRose and collaborators, see https://journals.aps.org/prc/abstract/10.1103/PhysRevC.106.024319, we can rewrite the Lipkin Hamiltonian for efficient adaptations to quantum computing.
We have

$$J_z = \sum_i^N j_z^i \qquad J_\pm = \sum_i^N j_\pm^i = \sum_i^N (j_x^i \pm i j_y^i),$$

with N being the number of particles. Additionally, since we have spin-1/2 fermions, the mapping to Pauli matrices follow

$$j_x^i = \frac{1}{2} X_i, \qquad j_y^i = \frac{1}{2} Y_i, \qquad j_z^i = \frac{1}{2} Z_i.$$

This means that we require N qubits to calculate properties of a N particle system, if no more symmetry reductions are considred.

Lipkin Hamiltonian for quantum computing

We rewrote our Hamiltonian as

$$
\begin{aligned}
H_0 &= \frac{\epsilon}{2}\sum_p Z_p, \\
H_1 &= \frac{1}{2}V\sum_{p<q}(X_pX_q - Y_pY_q), \\
H_2 &= \frac{1}{2}W\sum_{p<q}\left(X_pX_q + Y_pY_q\right).
\end{aligned}
$$

For two particles coupling to spin $J=1$ we have

$$
H^{N=2} = \frac{\epsilon}{2}\left(Z_1 + Z_2\right) + \frac{W+V}{2}X_1X_2 - \frac{W-V}{2}Y_1Y_2.
$$

For four fermions

Our Hamiltonian for $N = 4$ is

$$\begin{aligned} H^{N=4} =& \frac{\epsilon}{2}(Z_1 + Z_2 + Z_3 + Z_4) + \frac{W-V}{2}\Big(X_1X_2 + \\ & X_1X_3 + X_1X_4 + X_2X_3 + X_3X_4 + X_3X_4\Big) \\ & + \frac{W+V}{2}\Big(Y_1Y_2 + Y_1Y_3 + Y_1Y_4 + Y_2Y_3 \\ & + Y_3Y_4 + Y_3Y_4\Big). \end{aligned}$$

Note that well that a term like Z_1 reads $Z_1 \otimes I_2 \otimes I_3 \otimes I_4$ where the subscript points to qubit i and all matrices are 2×2 matrices.

Leaving out the W term

If we set $W = 0$ we can simplify the Hamiltonian. For this we note that if we only include the H_0 and H_1 terms, spins differing by ± 2 are the only possible non-diagonal couplings. Since H_0 is diagonal and single particle energies are degenerate, it does not break any symmetry. Instead of the $2J + 1$ spin projections, we simply have $J + 1$ relevant states. From `https://journals.aps.org/prc/abstract/10.1103/PhysRevC.106.024319`, the two-qubit Hamiltonian for $N = 4$ can be written as

$$H_{W=0}^{N=4} = \epsilon(Z_1 + Z_2) + \frac{\sqrt{6}}{2} V(X_1 + X_2 + Z_1 X_0 - X_1 Z_0).$$

This is computationally more efficient since we only need half of the qubits.

More details

Here we give some additional details behind the rewriting of the Lipkin Hamiltonian in term of Pauli matrices and the identity matrix. For the one-body term we have

$$H_0 = \epsilon J_z = \epsilon \sum_p j_z^p = \frac{\epsilon}{2} \sum_p Z_p$$

Moving on to the H_1 term, we need to expand the square of the $J_\pm$ operators. Starting with J_+ we find

$$\begin{aligned} J_+^2 &= (\sum_p j_x^p + \imath j_y^p)^2 = \sum_{pq} (j_x^p + \imath j_y^p)(j_x^q + \imath j_y^q) \\ &= \sum_{pq} \left(j_x^p j_x^q - j_y^p j_y^q + \imath j_y^p j_x^q + \imath j_x^p j_y^q \right) . \end{aligned}$$

Other manipulations

Similarly, the J_-^2 term yields

$$J_-^2 = \sum_{pq} (j_x^p j_x^q - j_y^p j_y^q - \imath j_y^p j_x^q - \imath j_x^p j_y^q).$$

Rewriting we get

$$\begin{aligned} J_+^2 + J_-^2 &= 2 \sum_{pq} j_x^p j_x^q - j_z^p j_z^q \\ &= 2 \sum_p (j_x^p)^2 - (j_y^p)^2 + 4 \sum_{p>q} (j_x^p j_x^q - j_y^p j_y^q) \\ &= \frac{1}{2} \sum_p (X_p^2 - Y_p^2) + \sum_{p>q} (X_p X_q - Y_p Y_q) \end{aligned}$$

Final expression for H_1 and H_2

The diagonal term will cancel out, since the Pauli matrices are involutory, resulting in

$$H_1 = \frac{1}{2} V \sum_{p>q} (X_p X_q - Y_p Y_q).$$

Following the same procedure as above, we have finally

$$H_2 = \frac{1}{2} W \sum_{p<q} (X_p X_q + Y_p Y_q).$$

Background material on VQE and transformations into a computational basis

The review of Tilly et al., see https://discovery.ucl.ac.uk/id/eprint/10157639/1/1-s2.0-S0370157322003118-main.pdf (Physics Reports **986**, 1 (2022)), gives examples on how to arrive at the equations (see section 5.2) listed in the tables here. The article gives also an excellent overview of the VQE method.
See also Hundt section 6.11.

Transforming Pauli and identity matrices

TABLE I. Operators for measuring Pauli strings of length 1 in the basis Z, and Pauli strings of length 2 in the basis $Z \otimes I$.

Pauli string $\mathcal{P} = \bigotimes_{n=0}^{N-1} P_i$	Operator $U_{\mathcal{P}}$
X	H
$Z \otimes I$	$I \otimes I$
$I \otimes Z$	SWAP
$X \otimes I$	$H \otimes I$
$I \otimes X$	$(H \otimes I)\,\mathrm{SWAP}$
$Y \otimes I$	$HS^\dagger \otimes I$
$I \otimes Y$	$\left(HS^\dagger \otimes I\right)\mathrm{SWAP}$
$Z \otimes Z$	$CX_{1,0}$
$X \otimes X$	$CX_{1,0}\,(H \otimes H)$
$Y \otimes Y$	$CX_{1,0}\left(HS^\dagger \otimes HS^\dagger\right)$
$Z \otimes X$	$CX_{1,0}\,(I \otimes H)$
$X \otimes Z$	$CX_{1,0}\,(I \otimes H)\,\mathrm{SWAP}$

Transforming Pauli and identity matrices, four particle case

TABLE II. Operators for measuring Pauli strings of length 4 in the basis $Z \otimes I \otimes I \otimes I$.

Pauli string $\mathcal{P} = \bigotimes_{n=0}^{N-1} P_i$	Operator $U_{\mathcal{P}}$
$Z \otimes I \otimes I \otimes I$	$I \otimes I \otimes I \otimes I$
$I \otimes Z \otimes I \otimes I$	$U_{IZ} \otimes I \otimes I$
$I \otimes I \otimes Z \otimes I$	$U_{IZII}\,(I \otimes U_{IZ} \otimes I)$
$I \otimes I \otimes I \otimes Z$	$U_{IIZI}\,(I \otimes I \otimes U_{IZ})$
$Z \otimes I \otimes Z \otimes I$	$(CX_{1,0} \otimes I \otimes I)\,(\mathrm{SWAP} \otimes I \otimes I)$
$X \otimes X \otimes I \otimes I$	$U_{XX} \otimes I \otimes I$
$X \otimes I \otimes X \otimes I$	$U_{ZIZI}\,(U_{XI} \otimes U_{XI})$
$X \otimes I \otimes I \otimes X$	$U_{ZIZI}\,(U_{XI} \otimes U_{IX})$
$I \otimes X \otimes X \otimes I$	$U_{ZIZI}\,(U_{IX} \otimes U_{XI})$
$I \otimes X \otimes I \otimes X$	$U_{ZIZI}\,(U_{IX} \otimes U_{IX})$
$I \otimes I \otimes X \otimes X$	$U_{IIZI}\,(I \otimes I \otimes U_{XX})$
$Y \otimes Y \otimes I \otimes I$	$U_{YY} \otimes I \otimes I$
$Y \otimes I \otimes Y \otimes I$	$U_{ZIZI}\,(U_{YI} \otimes U_{YI})$
$Y \otimes I \otimes I \otimes Y$	$U_{ZIZI}\,(U_{YI} \otimes U_{IY})$
$I \otimes Y \otimes Y \otimes I$	$U_{ZIZI}\,(U_{IY} \otimes U_{YI})$
$I \otimes Y \otimes I \otimes Y$	$U_{ZIZI}\,(U_{IY} \otimes U_{IY})$
$I \otimes I \otimes Y \otimes Y$	$U_{IIZI}\,(I \otimes I \otimes U_{YY})$

Plans for the week March 18-22

1. Discussion of project 1 and possible paths for project 2
2. Introduction to Quantum Fourier transforms